Business Report

1. Summary of Predictive Insights

The predictive model was developed to identify customers who are likely to become delinquent (miss loan or credit repayments). After cleaning the data, handling missing values, engineering new features, balancing the dataset using SMOTE, and training a Logistic Regression model, the following business insights were obtained.

Model Performance

Metric	Result	Business Interpretation	
Accuracy	**58%**	The model correctly classified about half of the customers.	
ROC-AUC	**0.46**	The model has limited ability to distinguish between delinquent and non-delinquent customers. Further model improvement is recommended before production use.	
Recall (Delinquent Customers)	**42%**	The model identified only a small portion of customers who actually became delinquent.	

Key Predictive Factors

The model identified the following variables as the strongest indicators of customer delinquency:

- **High Credit Utilization** → Customers using a large portion of their available credit are more likely to become delinquent.
- **High Debt-to-Income Ratio** → Customers whose debts consume a large share of their income have a higher repayment risk.
- **Continuous Missed Payments** → Customers with repeated missed payments over previous months have the highest likelihood of future delinquency.
- **Low Account Tenure** → Newer customers are more likely to default than long-term customers.
- **Employment Status** → Unemployed customers show significantly higher delinquency rates.
- **Credit Card Type** → Business and Student card holders exhibit relatively higher delinquency compared to Platinum or Standard card holders.

Business Value

The predictive model helps the bank:

- Identify high-risk customers before they default.
- Prioritize early intervention for risky customers.
- Reduce loan losses through proactive risk management.
- Support better lending and credit monitoring decisions.

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2. Recommendation Framework

Principle	Recommendation
Specific	Focus on customers with high credit utilization, high debt-to-income ratio, repeated missed payments, and short account tenure.
Measurable	Reduce delinquent accounts by **10–15%** and improve recall of delinquent customer detection to **above 60%** within the next model improvement cycle.
Achievable	Implement automated monthly risk scoring, payment reminders, credit limit reviews, and financial counseling for high-risk customers.
Relevant	These actions directly support the bank's objective of reducing credit losses while improving customer financial health.
Time-Bound	Review customer risk scores every month and evaluate model performance every quarter. Retrain the model using new customer data every **3–6 months**.

Priority Business Actions

High Priority

- Monitor customers with repeated missed payments.
- Send early payment reminders.
- Review credit limits for customers with very high credit utilization.

Medium Priority

- Provide financial education to customers with high debt-to-income ratios.
- Closely monitor newly onboarded customers during their first year.

Long-Term Priority

- Collect additional customer information.
- Improve data quality.

- Experiment with more advanced machine learning models to improve prediction accuracy.

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3. Ethical and Responsible AI Considerations

Fairness

The AI model should treat every customer fairly. Decisions should not unfairly favor or disadvantage customers because of personal characteristics such as age, gender, or location.

Transparency

The bank should be able to explain why a customer is considered high risk. Customers and employees should understand the main reasons behind the prediction.

Privacy

Customer information must be kept secure and used only for approved business purposes. Personal financial data should never be shared without permission.

Human Decision Making

The AI model should **support** decision-making, not replace it. Final decisions about loans or customer actions should always be reviewed by bank employees.

Continuous Monitoring

Customer behavior changes over time. The model should be regularly monitored and retrained using recent data to maintain accuracy and fairness.

Data Quality

Predictions are only as good as the data used. Missing values, incorrect records, and inconsistent information should be cleaned before training the model.

Final Business Conclusion

The predictive model successfully identifies important factors associated with customer delinquency, such as **credit utilization, debt-to-income ratio, missed payment history, employment status, and account tenure**. Although the current Logistic Regression model provides useful business insights, its predictive performance is relatively low (Accuracy **58%**, ROC-AUC **0.46**). Therefore, the model should be treated as a decision-support tool rather than a fully automated decision system. By improving data quality, using more advanced machine learning algorithms, and regularly monitoring model performance, the bank can enhance its ability to identify high-risk customers early, reduce financial losses, and make more informed lending decisions.